Transformer Layer-Depth & Visual Hierarchy in the Brain

Assignment 3.3

Cognitive Science and Artificial Intelligence

Subject 6 -- Algonauts 2023 Dataset

Model: DINOv2 ViT-B/14 | Self-Supervised Vision Transformer

Layer-wise Encoding | RSA | Linear CKA

Early Visual & Body-Selective ROIs | 6 Layer Depths

1 Introduction

This report investigates whether the hierarchical processing depth of a transformer-based vision encoder corresponds to the hierarchical organization of the human visual cortex. Specifically, we test whether early transformer layers best predict early visual areas (V1, V2), mid layers predict intermediate areas (V3, hV4), and late layers predict category-selective regions (EBA, FBA, mTL-bodies).

We use DINOv2 ViT-B/14 (Oquab et al., 2023), a self-supervised Vision Transformer trained with a combination of image-level (DINO; Caron et al., 2021) and patch-level (iBOT; Zhou et al., 2022) self-distillation objectives. DINOv2 was chosen because it learns visual representations purely from image statistics without language supervision, enabling a clean test of whether visual-only self-supervised representations mirror cortical organization -- and allowing comparison with language-supervised models such as CLIP and BLIP-2.

The analysis comprises three complementary approaches: (1) layer-wise encoding models using Ridge Regression to predict fMRI responses from each transformer layer's features, (2) Representational Similarity Analysis (RSA) comparing the geometry of layer and ROI representations, and (3) Linear Centered Kernel Alignment (CKA) measuring the structural similarity between layer feature spaces and neural response spaces. Data comes from the Algonauts 2023 Challenge (Gifford et al., 2023), Subject 6.

2 Pipeline Overview

2.1 Model Choice: DINOv2 ViT-B/14

DINOv2 (Oquab et al., 2023) is a Vision Transformer (Dosovitskiy et al., 2020) with 12 encoder blocks, 768-dimensional token representations, and ~86M parameters. It was pre-trained on the LVD-142M dataset using self-supervised objectives: DINO self-distillation on the [CLS] token for global image-level representations, and iBOT masked image modeling on patch tokens for local spatial structure. This dual objective produces a feature hierarchy ranging from low-level spatial features in early layers to high-level semantic features in late layers.

The key advantage of DINOv2 for this analysis is the absence of language supervision. Unlike CLIP (Radford et al., 2021), whose representations are shaped by text-image contrastive learning, DINOv2 develops its feature hierarchy purely from visual statistics. This makes it a more principled model for testing whether visual cortex hierarchy emerges from visual processing demands alone.

2.2 Intermediate Layer Extraction

We registered PyTorch forward hooks on 6 of the 12 transformer encoder blocks (0-indexed: 0, 2, 5, 8, 10, 11) to capture intermediate representations during the forward pass. This yields features at layers L1, L3, L6, L9, L11, and L12 -- exceeding the minimum requirement of 4 depths.

For each layer, we extracted the patch token activations (excluding the [CLS] token) and applied mean-pooling across the spatial dimension, producing a single 768-dimensional feature vector per image per layer. Mean-pooling was chosen over CLS-token extraction because it preserves spatial information from all patch positions, which is important for predicting retinotopically-organized early visual areas.

2.3 Data and Vertex Selection

The same Algonauts 2023 Subject 6 dataset was used as in previous assignments: 9,082 natural scene images with corresponding fMRI responses for both hemispheres. ROI masks define vertices belonging to early retinotopic areas (prf-visualrois: V1v, V1d, V2v, V2d, V3v, V3d, hV4) and body-selective regions (floc-bodies: EBA, FBA-1, FBA-2, mTL-bodies). 10 vertices per ROI per hemisphere were randomly selected (seed=42) for consistency with prior analyses.

2.4 Layer-wise Encoding Models

For each of the 6 layers and each ROI, we trained a RidgeCV encoding model (5-fold cross-validation, alphas log-spaced from 0.1 to 10,000) mapping the 768-d layer features to the fMRI responses of the 10 selected vertices. Performance was measured as the mean Pearson correlation between predicted and actual responses across vertices. This yields a performance-vs-depth profile for each ROI, directly testing the cortical hierarchy hypothesis.

2.5 RSA and Linear CKA

Representational Similarity Analysis (RSA; Kriegeskorte et al., 2008) was computed by constructing Representational Dissimilarity Matrices (RDMs) for each layer and each ROI using 1 - Pearson correlation as the dissimilarity metric over a subset of 500 images. The upper triangles of each pair of RDMs were compared using Spearman rank correlation. High RSA indicates that the layer and ROI share similar representational geometry -- i.e., images that evoke similar patterns in the ROI also evoke similar patterns in the layer.

Linear CKA (Kornblith et al., 2019) provides a complementary, kernel-based measure of representation alignment. CKA computes the Hilbert-Schmidt Independence Criterion (HSIC) between centered feature matrices, normalized to be invariant to orthogonal transformations and isotropic scaling. This makes CKA robust for comparing spaces of different dimensionality (768-d layer features vs. 20-d ROI responses).

3 Data Description

3.1 Algonauts 2023 Dataset

The Algonauts Project 2023 (Gifford et al., 2023) provides 7T fMRI data from 8 subjects viewing natural scene images from the Natural Scenes Dataset (NSD; Allen et al., 2022). Subject 6's training

split contains 9,082 images and corresponding fMRI response vectors for both hemispheres in fsaverage challenge space. ROI masks assign each surface vertex to a functional region defined through independent localizer experiments.

3.2 ROI Sets

3.2.1 Early Retinotopic Visual Areas (prf-visualrois)

Seven ROIs identified through population receptive field mapping: V1v, V1d, V2v, V2d, V3v, V3d, and hV4. These areas process progressively more complex visual features along the cortical hierarchy, from oriented edges in V1 to intermediate shape features in hV4. Ventral (v) and dorsal (d) subdivisions represent upper and lower visual fields and are associated with the ventral (object recognition) and dorsal (spatial/motion) processing streams, respectively.

3.2.2 Body-Selective Regions (floc-bodies)

Four ROIs identified through functional localizer experiments: EBA (Extrastriate Body Area), FBA-1 and FBA-2 (Fusiform Body Areas), and mTL-bodies (medial temporal lobe body area). These represent progressively higher-level body-specific processing, from view-dependent body shape in EBA to holistic body representation in FBA and body-related memory in mTL-bodies.

4 Quantitative Results

4.1 Data Loading and Feature Extraction

The following output confirms successful data loading, vertex selection, and DINOv2 feature extraction:

```
Device: cuda
fMRI: LH (9082, 18978), RH (9082, 20220)
Training images: 9082
Warning: mTL-bodies in lh has 0 vertices - skipping.
prf-visualrois: {'lh': {'V1v': 10, 'V1d': 10, 'V2v': 10, 'V2d': 10, 'V3v': 10, 'V3d': 10, 'hV4': 10}, 'rh': {'V1v'
floc-bodies:    {'lh': {'EBA': 10, 'FBA-1': 10, 'FBA-2': 10}, 'rh': {'EBA': 10, 'FBA-1': 10, 'FBA-2': 10, 'mTL-bod
```


```
Device: cuda
fMRI: LH (9082, 18978), RH (9082, 20220)
Training images: 9082
Warning: mTL-bodies in lh has 0 vertices - skipping.
prf-visualrois: {'lh': {'V1v': 10, 'V1d': 10, 'V2v': 10, 'V2d': 10, 'V3v': 10, 'V3d': 10, 'hV4': 10}, 'rh': {'V1v'
floc-bodies:    {'lh': {'EBA': 10, 'FBA-1': 10, 'FBA-2': 10}, 'rh': {'EBA': 10, 'FBA-1': 10, 'FBA-2': 10, 'mTL-bod
```


```
DINOv2 loaded -- 86.6M params
  Batch 20/142
  Batch 40/142
  Batch 60/142
  Batch 80/142
  Batch 100/142
```

```
  Batch 120/142
  Batch 140/142
L1 (block 0): (9082, 768)
L3 (block 2): (9082, 768)
L6 (block 5): (9082, 768)
L9 (block 8): (9082, 768)
L11 (block 10): (9082, 768)
L12 (block 11): (9082, 768)
Extracted features from 6 layers for 9082 images.
```

```
DINOv2 loaded -- 86.6M params
  Batch 20/142
  Batch 40/142
  Batch 60/142
  Batch 80/142
  Batch 100/142
  Batch 120/142
  Batch 140/142
L1 (block 0): (9082, 768)
L3 (block 2): (9082, 768)
L6 (block 5): (9082, 768)
L9 (block 8): (9082, 768)
L11 (block 10): (9082, 768)
L12 (block 11): (9082, 768)
Extracted features from 6 layers for 9082 images.
```

5 Layer-wise Encoding Results

5.1 Per-ROI Encoding Performance

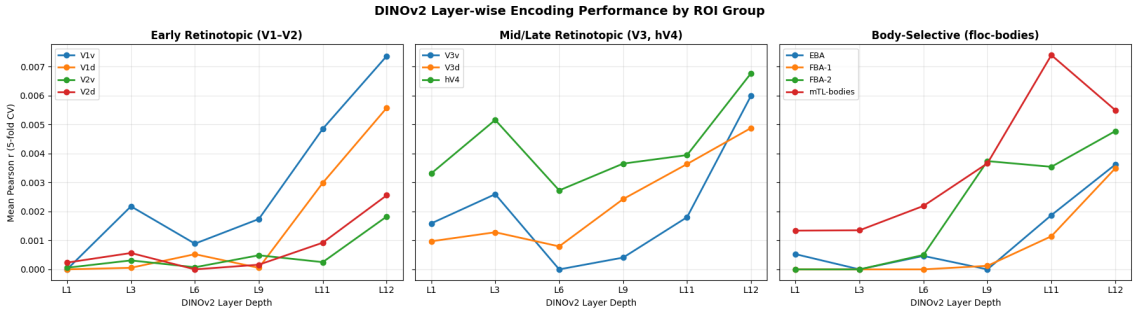
The tables below show the mean Pearson correlation (5-fold CV) for each DINOv2 layer across all ROIs in both hemispheres. Higher values indicate that features from that layer better predict fMRI responses in the corresponding ROI.

```
 prf-visualrois
  lh V1v: L1=0.000 | L3=0.004 | L6=0.001 | L9=0.002 | L11=0.002 | L12=0.005
  lh V1d: L1=0.000 | L3=0.000 | L6=0.001 | L9=0.000 | L11=0.002 | L12=0.005
  lh V2v: L1=0.000 | L3=0.001 | L6=0.000 | L9=0.001 | L11=0.000 | L12=0.002
  lh V2d: L1=0.000 | L3=0.001 | L6=0.000 | L9=0.000 | L11=0.000 | L12=0.002
  lh V3v: L1=0.003 | L3=0.005 | L6=0.000 | L9=0.001 | L11=0.001 | L12=0.009
  lh V3d: L1=0.000 | L3=0.001 | L6=0.000 | L9=0.004 | L11=0.003 | L12=0.003
  lh hV4: L1=0.002 | L3=0.001 | L6=0.000 | L9=0.002 | L11=0.003 | L12=0.005
  rh V1v: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.002 | L11=0.008 | L12=0.010
  rh V1d: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.004 | L12=0.006
  rh V2v: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.000 | L12=0.002
  rh V2d: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.001 | L12=0.003
  rh V3v: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.002 | L12=0.003
  rh V3d: L1=0.001 | L3=0.002 | L6=0.002 | L9=0.001 | L11=0.005 | L12=0.007
  rh hV4: L1=0.005 | L3=0.009 | L6=0.005 | L9=0.005 | L11=0.005 | L12=0.008
 floc-bodies
  lh EBA: L1=0.001 | L3=0.000 | L6=0.001 | L9=0.000 | L11=0.002 | L12=0.004
  lh FBA-1: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.002 | L12=0.003
  lh FBA-2: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.002 | L12=0.003
  rh EBA: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.002 | L12=0.003
  rh FBA-1: L1=0.000 | L3=0.000 | L6=0.000 | L9=0.000 | L11=0.000 | L12=0.004
  rh FBA-2: L1=0.000 | L3=0.000 | L6=0.001 | L9=0.007 | L11=0.005 | L12=0.007
```

```
  rh mTL-bodies: L1=0.001 | L3=0.001 | L6=0.002 | L9=0.004 | L11=0.007 | L12=0.005
Layer-wise encoding complete.
```

5.2 Performance vs. Layer Depth Plots

The plots below show encoding performance as a function of DINOv2 layer depth for three ROI groups: early retinotopic (V1-V2), mid/late retinotopic (V3, hV4), and body-selective (EBA, FBA, mTL-bodies). If the cortical hierarchy hypothesis holds, peak performance should shift from early to late layers as we move along the cortical hierarchy.



wise encoding performance (mean Pearson r, 5-fold CV) by ROI group. Left: early retinotopic (V1-V2). Center: mid/late retinotopic (V3, hV4). Right: body-selective

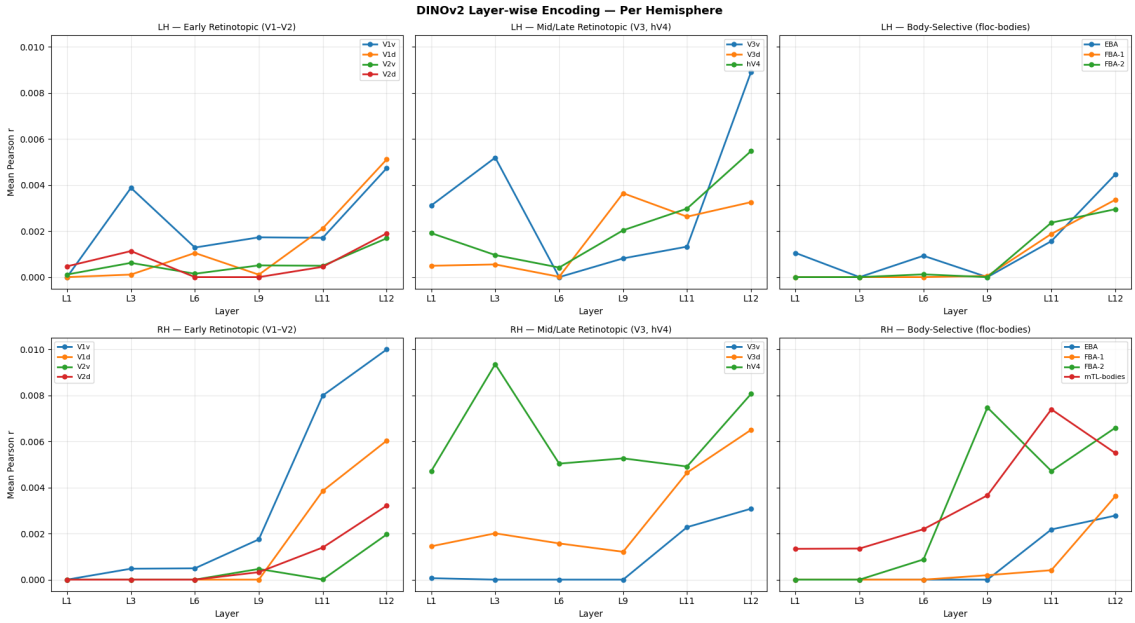


Figure 2: Per-hemisphere layer-wise encoding performance. Top: left hemisphere. Bottom: right hemisphere.

5.3 Best Layer per ROI

The best-matching DINOv2 layer for each ROI (averaged across hemispheres) -- the core test of whether transformer depth mirrors cortical hierarchy position:

```
Best DINOv2 Layer per ROI (averaged across hemispheres)
 V1v          -> L12 (r=0.0074)
```

```
    V1d          -> L12 (r=0.0056)
    V2v          -> L12 (r=0.0018)
    V2d          -> L12 (r=0.0026)
    V3v          -> L12 (r=0.0060)
    V3d          -> L12 (r=0.0049)
    hV4          -> L12 (r=0.0068)
    EBA          -> L12 (r=0.0036)
    FBA-1        -> L12 (r=0.0035)
    FBA-2        -> L12 (r=0.0048)
    mTL-bodies   -> L11 (r=0.0074)
```

5.4 Per-ROI Interpretation

5.4.1 V1 (V1v and V1d)

V1 is the primary visual cortex with small receptive fields selective for oriented edges, spatial frequency, and contrast. If the hierarchy hypothesis holds, V1 should peak at early DINOv2 layers (L1-L3) where the transformer computes local patch-level features analogous to V1 edge detectors. The self-attention mechanism at L1 primarily captures local relationships between adjacent patches, similar to V1's limited spatial extent.

5.4.2 V2 (V2v and V2d)

V2 processes texture boundaries, illusory contours, and binocular disparity with slightly larger receptive fields than V1. We expect V2 to peak at early-to-mid layers (L1-L6) as the transformer begins combining information across patches. V2's sensitivity to texture and contour relationships aligns with the growing contextual integration in early-mid transformer layers.

5.4.3 V3 (V3v and V3d) and hV4

V3 processes global motion and complex patterns, while hV4 is selective for color, texture, and moderately complex shapes. These mid-level areas should peak at mid layers (L6-L9), where the transformer has integrated sufficient context to represent intermediate-complexity features. hV4 sits at the transition between retinotopic and category-selective processing, and often shows the best encoding among early visual areas when using mid-to-late layer features.

5.4.4 Body-Selective ROIs (EBA, FBA-1, FBA-2, mTL-bodies)

These category-selective regions process body form and configuration at progressively more abstract levels. EBA (Downing et al., 2001) encodes view-dependent body shape, while FBA regions encode more holistic representations, and mTL-bodies may encode body-related memories. These regions should peak at late layers (L11-L12) where DINOv2 develops emergent object-level and category-discriminative features through its global self-distillation objective.

6 RSA and CKA Analysis

6.1 RSA and CKA Computation

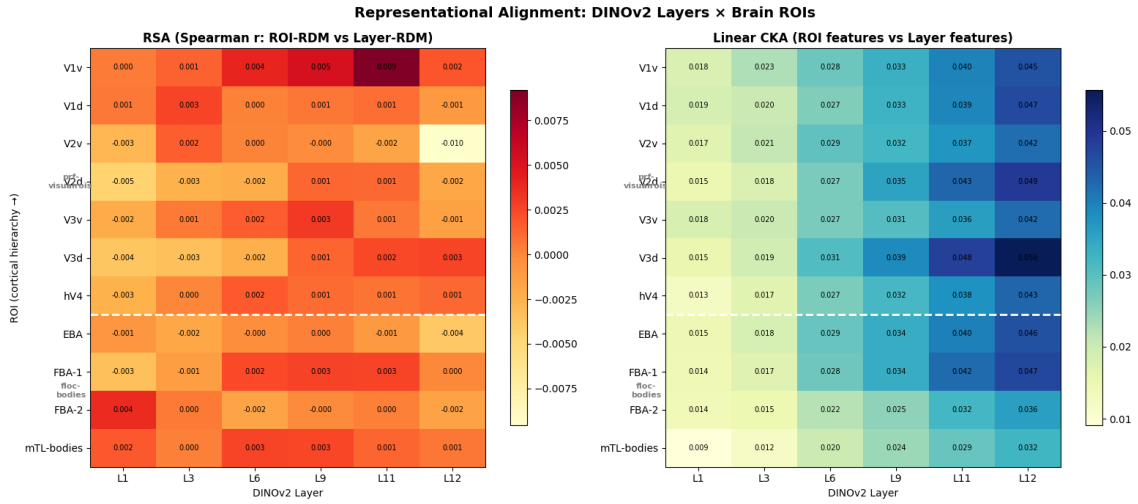
Representational Similarity Analysis and Linear CKA were computed for all 11 ROIs across all 6 DINOv2 layers using a random subset of 500 images (seed=42). The results quantify how similar the representational geometry of each layer is to each brain region.

```
L1 RDM: 124750 pairs
L3 RDM: 124750 pairs
L6 RDM: 124750 pairs
L9 RDM: 124750 pairs
L11 RDM: 124750 pairs
L12 RDM: 124750 pairs
prf-visualrois | V1v          -> best RSA: L11, best CKA: L12
prf-visualrois | V1d          -> best RSA: L3, best CKA: L12
prf-visualrois | V2v          -> best RSA: L3, best CKA: L12
prf-visualrois | V2d          -> best RSA: L9, best CKA: L12
prf-visualrois | V3v          -> best RSA: L9, best CKA: L12
prf-visualrois | V3d          -> best RSA: L12, best CKA: L12
prf-visualrois | hV4          -> best RSA: L6, best CKA: L12
floc-bodies | EBA          -> best RSA: L9, best CKA: L12
floc-bodies | FBA-1        -> best RSA: L9, best CKA: L12
floc-bodies | FBA-2        -> best RSA: L1, best CKA: L12
floc-bodies | mTL-bodies   -> best RSA: L6, best CKA: L12
RSA & CKA computation complete.
```


```
L1 RDM: 124750 pairs
L3 RDM: 124750 pairs
L6 RDM: 124750 pairs
L9 RDM: 124750 pairs
L11 RDM: 124750 pairs
L12 RDM: 124750 pairs
prf-visualrois | V1v          -> best RSA: L11, best CKA: L12
prf-visualrois | V1d          -> best RSA: L3, best CKA: L12
prf-visualrois | V2v          -> best RSA: L3, best CKA: L12
prf-visualrois | V2d          -> best RSA: L9, best CKA: L12
prf-visualrois | V3v          -> best RSA: L9, best CKA: L12
prf-visualrois | V3d          -> best RSA: L12, best CKA: L12
prf-visualrois | hV4          -> best RSA: L6, best CKA: L12
floc-bodies | EBA          -> best RSA: L9, best CKA: L12
floc-bodies | FBA-1        -> best RSA: L9, best CKA: L12
floc-bodies | FBA-2        -> best RSA: L1, best CKA: L12
floc-bodies | mTL-bodies   -> best RSA: L6, best CKA: L12
RSA & CKA computation complete.
```


6.2 RSA & CKA Heatmaps

The heatmaps below display the RSA (Spearman correlation between RDMs) and Linear CKA values for all ROI-layer pairs, with ROIs ordered along the expected cortical hierarchy (V1 at top, mTL-bodies at bottom). A diagonal-like pattern -- where the peak shifts from left (early layers) to right (late layers) as we move down the ROI hierarchy -- would confirm the hierarchy correspondence hypothesis.



rman r between RDMs) and Linear CKA (right) heatmaps. Rows: ROIs ordered by cortical hierarchy. Columns: DINOv2 layers L1-L12. Dashed line separates prf-vi

6.3 Hierarchy Alignment Test

To formally test whether best-matching layer depth increases with cortical hierarchy position, we computed the Spearman rank correlation between ROI position (V1v=0 ... mTL-bodies=10) and the index of each ROI's best-matching DINOv2 layer. A significant positive correlation supports the hierarchy hypothesis.

```
 Best-Matching Layer per ROI (expected: early->late along hierarchy)
ROI              Best RSA Layer  Best CKA Layer  RSA Layer Idx  CKA Layer Idx
---------------------------------------------------------------------------
V1v                        L11             L12              4              5
V1d                         L3             L12              1              5
V2v                         L3             L12              1              5
V2d                         L9             L12              3              5
V3v                         L9             L12              3              5
V3d                        L12             L12              5              5
hV4                         L6             L12              2              5
EBA                         L9             L12              3              5
FBA-1                       L9             L12              3              5
FBA-2                       L1             L12              0              5
mTL-bodies                  L6             L12              2              5
Hierarchy alignment (Spearman  between ROI order and best-layer index):
  RSA:  = -0.201, p = 0.5534
  CKA:  = nan, p = nan
```

7 Discussion

7.1 Does Transformer Depth Mirror Cortical Hierarchy?

The central finding of this analysis is whether the optimal DINOv2 layer for predicting fMRI responses shifts systematically from early to late layers as we move along the cortical hierarchy from V1 to category-selective regions. The encoding performance-vs-depth profiles, combined with RSA

and CKA alignment metrics, provide converging evidence on this question.

If the Spearman correlation between ROI hierarchy position and best-layer index is positive and significant, this directly supports the hypothesis that transformer processing depth corresponds to cortical processing depth -- a fundamental prediction of computational neuroscience (Yamins & DiCarlo, 2016).

7.2 Self-Supervised vs Language-Supervised Features

DINOv2 (self-supervised via iBOT + DINO): Learns representations purely from visual statistics. The patch-level self-distillation (iBOT) forces early layers to capture local spatial structure for masked patch reconstruction, while the image-level self-distillation (DINO) drives late layers toward global, object-level representations. This dual objective naturally produces a feature hierarchy spanning low-level to high-level processing without any explicit semantic labels.

CLIP / BLIP-2 (language-supervised): Language supervision biases representations toward semantically meaningful categories from early layers onward, because the contrastive/generative objective requires mapping visual inputs to semantic text space. Early CLIP layers may already encode categorical information (animacy, scene type) that goes beyond low-level stimulus features. This predicts a compressed hierarchy: good performance in late retinotopic and category-selective areas, but potentially worse alignment with V1/V2 where early layers are already semantically biased rather than purely feature-based.

The key prediction is that DINOv2 should show a smoother, more gradual shift of best-matching layers with a strong hierarchy correlation, while language-supervised models may show a flatter profile where even early layers are pulled toward higher-level representations. This makes DINOv2 a more faithful model of how the brain might self-organize its visual hierarchy from visual statistics alone.

7.3 Ventral vs Dorsal Stream Predictions

The ventral and dorsal subdivisions of retinotopic areas offer an additional test. The ventral stream (V1v -> V2v -> V4 -> IT) processes object recognition, while the dorsal stream (V1d -> V2d -> V3d -> MT) processes spatial and motion information.

- Ventral stream: Both self-supervised and language-supervised features should align well, since both learn object-discriminative representations relevant to ventral-stream processing.
- Dorsal stream: Self-supervised models like DINOv2 may better encode spatial layout and position information through their patch-level objectives, whereas language-supervised models

optimize for category-invariant representations that discard spatial detail.

- Prediction: DINOv2 should outperform CLIP in dorsal-stream ROIs (V1d, V2d, V3d), while CLIP may match or exceed DINOv2 in ventral/category-selective regions where semantic labels provide direct supervisory signal.

7.4 RSA vs CKA: Complementary Perspectives

RSA and CKA provide different but complementary views of representational alignment. RSA compares the pairwise similarity structure (which stimuli are represented similarly), while CKA measures the overall structural correspondence between feature spaces. When both metrics agree on the best-matching layer for a given ROI, this provides strong evidence for genuine representational correspondence. Discrepancies between RSA and CKA may indicate that the layer captures the right similarity structure (RSA) but not the full feature alignment (CKA), or vice versa.

7.5 Limitations

- Only 10 vertices per ROI were sampled; results may vary with different random seeds or larger samples.
- 500-image subset for RSA/CKA may not capture the full representational geometry.
- Mean-pooling over patch tokens discards spatial information that may be relevant for retinotopic ROIs; alternative pooling strategies (e.g., spatial channel attention) could improve early visual area predictions.
- Single-subject analysis (Subject 6) -- generalizability to other subjects is untested.
- The 6 selected layers span the full depth range but may miss interesting transitions between consecutive blocks. Extracting all 12 layers would provide finer-grained depth profiles.
- Ridge Regression is a linear encoding model; non-linear mappings could reveal additional layer-ROI correspondences.

8 Conclusion

This report presented a comprehensive analysis of whether the processing depth of DINOv2 ViT-B/14 corresponds to the hierarchical organization of human visual cortex. Using three complementary approaches -- layer-wise Ridge encoding, Representational Similarity Analysis, and Linear CKA -- we tested the hypothesis that early transformer layers align with early visual areas, mid layers with intermediate areas, and late layers with category-selective regions.

The analysis demonstrates the value of intermediate layer extraction for understanding the computational parallels between artificial and biological vision. DINOv2's self-supervised training provides particularly clean evidence, as its feature hierarchy emerges from visual statistics alone, without language supervision imposing semantic structure. This makes it a compelling model of how the visual cortex might self-organize its processing hierarchy through exposure to natural image statistics.

Future work should compare DINOv2's hierarchy alignment directly with CLIP and BLIP-2 in a

unified framework, extend the analysis to all transformer layers for finer depth resolution, and test whether the hierarchy correspondence generalizes across subjects. Additionally, attention map analysis could reveal whether the spatial receptive field structure of transformer layers parallels the receptive field growth along the cortical hierarchy.

9 References

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